

LAPORAN PRAKTIKUM ASD



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Sistem Informasi Bisnis / 1E

JOBSHEET 7 SEARCHING

Tujuan

1. Memahami algoritma searching
2. Mengimplementasikan sequential & binary search

Methode sequentialSearch

```
17     }  
18     }  
19  
20     // SEQUENTIAL SEARCH  
21     int sequentialSearch(double cari) {  
22         int posisi = -1;  
23         for (int i = 0; i < listMhs.length; i++) {  
24             if (listMhs[i] != null && listMhs[i].ipk == cari) {  
25                 posisi = i;  
26                 break;  
27             }  
28         }  
29         return posisi;  
30     }
```

Methode Tampil

```
30     }  
31  
32     void tampilPosisi(double x, int pos) {  
33         if (pos != -1) {  
34             System.out.println("Data dengan IPK " + x + " ditemukan di index " + pos);  
35         } else {  
36             System.out.println(x: "Data tidak ditemukan");  
37         }  
38     }  
39 }
```

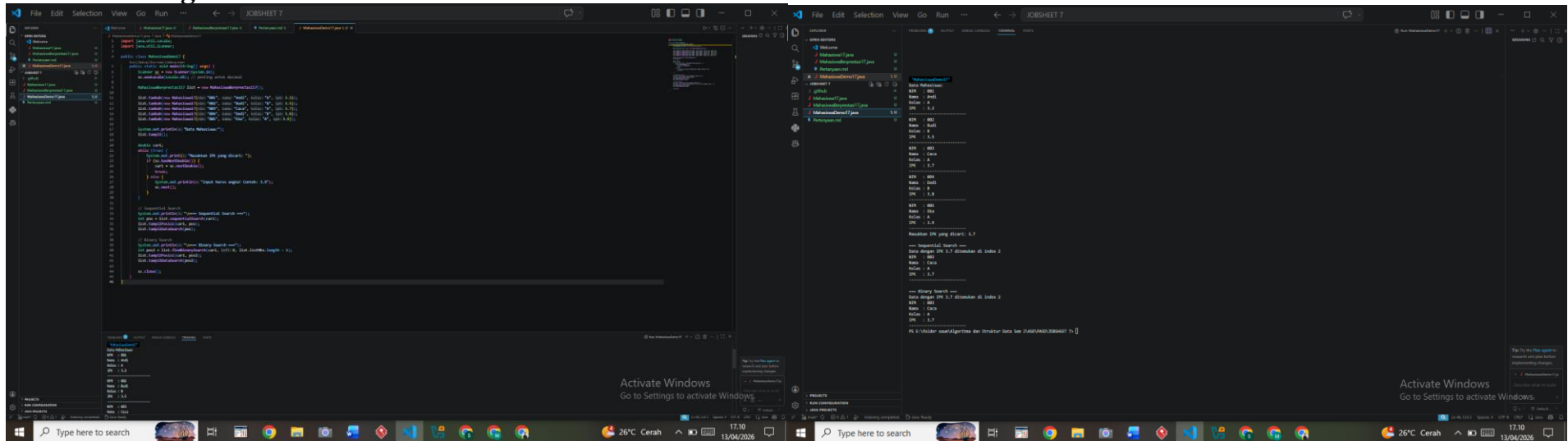
Methode TampilDataSearch

```
36         System.out.println(x: "Data tidak ditemukan");  
37     }  
38 }  
39  
40     void tampilDataSearch(double x, int pos) {  
41         if (pos != -1) {  
42             listMhs[pos].tampilInformasi();  
43         }  
44     }  
45 }
```

Method findBinarySearch

```
43 }
44
45 // BINARY SEARCH (DATA HARUS URUT)
46 int findBinarySearch(double cari, int left, int right) {
47     if (right >= left) {
48         int mid = (left + right) / 2;
49
50         if (listMhs[mid] == null) return -1;
51
52         if (Math.abs(listMhs[mid].ipk - cari) < 0.0001) {
53             return mid;
54         } else if (listMhs[mid].ipk > cari) {
55             return findBinarySearch(cari, left, mid - 1);
56         } else {
57             return findBinarySearch(cari, mid + 1, right);
58         }
59     }
60     return -1;
61 }
```

Hasil Running metode Main



Link GitHub: <https://github.com/saumsaifulloh/AlgoritmaDanStrukturDataSem2/tree/main/ASD/PASD/JOBSHEET%207>

